The aim of this study was to develop a computational model to estimate the annual salmon migration in the Tornionjoki River. The model is based on the Atlantic salmon (*Salmo salar*) life cycle in Tornionjoki, as described in Tuomas Manelius's Master's thesis. Salmon fishing tourism in the Tornionjoki River is estimated to be worth around ten million euros.

Atlantic salmon are an anadromous species whose life cycle begins in freshwater rivers, after which juveniles migrate to the sea, where they feed before returning to their natal river to spawn. In the Tornionjoki River, salmon typically spend 2–5 years in freshwater before migrating to the sea, where they remain for 1–4 years while experiencing mortality from predation, including seals, and fishing. After spawning, some individuals return to the sea, while others remain in the river over winter.

A simplified state-based model was constructed from this life cycle, incorporating fishing mortality and predation by seals. A computational model based on the Leslie population model was implemented in the Rust programming language and calibrated using observed data on Tornionjoki salmon migration.

The resulting model reproduces observed migration patterns from 2017 to 2022 and captures the overall fluctuations in salmon abundance. It was then used to project migration trends up to 2030. The model predicts a lag between strong and weak salmon years, predicting high migration years in 2027 and 2028, followed by a decline toward the end of the decade due to recent weak salmon years.

A sensitivity analysis was conducted by varying model inputs using a nearly orthogonal Latin hypercube design. The results indicate that the model parameters are tightly constrained, and that changes in fecundity can lead to either exponential growth or population decline.

The model could be further improved by incorporating additional biological structure, such as sex-specific differences.